

Class 10: Halloween Mini Project

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As it is nearly Halloween and the half way point in the quarter let's do a mini project to help us figure out the best candy!

Our data comes from the 538 website and is available as a CSV file>:

```
library(readr)
candy <- read.csv("https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power")
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0
	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

```
library(flextable)
flextable::flextable(head(candy))
```

chocolate	fruity	caramel	peanut	almond	nougat	crisped	rice	wafer	hard	bar	pluribus s
1	0	1	0	0	0	1	0	0	1	0	
1	0	0	0	0	1	0	0	0	1	0	
0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	
0	1	0	0	0	0	0	0	0	0	0	
1	0	0	1	0	0	0	0	0	1	0	

Q1. How many different candy types are in this dataset?

```
library(tidyverse)
candy %>%
  nrow()
```

[1] 85

Q2. How many fruity candy types are in the dataset?

```
candy$fruity %>%
  sum()
```

[1] 38

Q3. What is your favorite candy in the dataset and what is it's winpercent value?

My favourite candy is Trolli so...

```
candy["Trolli Sour Bites", ]$winpercent
```

[1] 47.17323

Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

Quick overview of the data set

```
library(skimr)
skimr::skim(candy)
```

Table 2: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Winpercent is on a different scale than the rest of variables, because it does not range between 0 and 1. Although, winpercent is a percentage which means that it also ranges between 0 and 1. The variables that only have 0 or 1s are binaries, because they only represent a virtual yes or no.

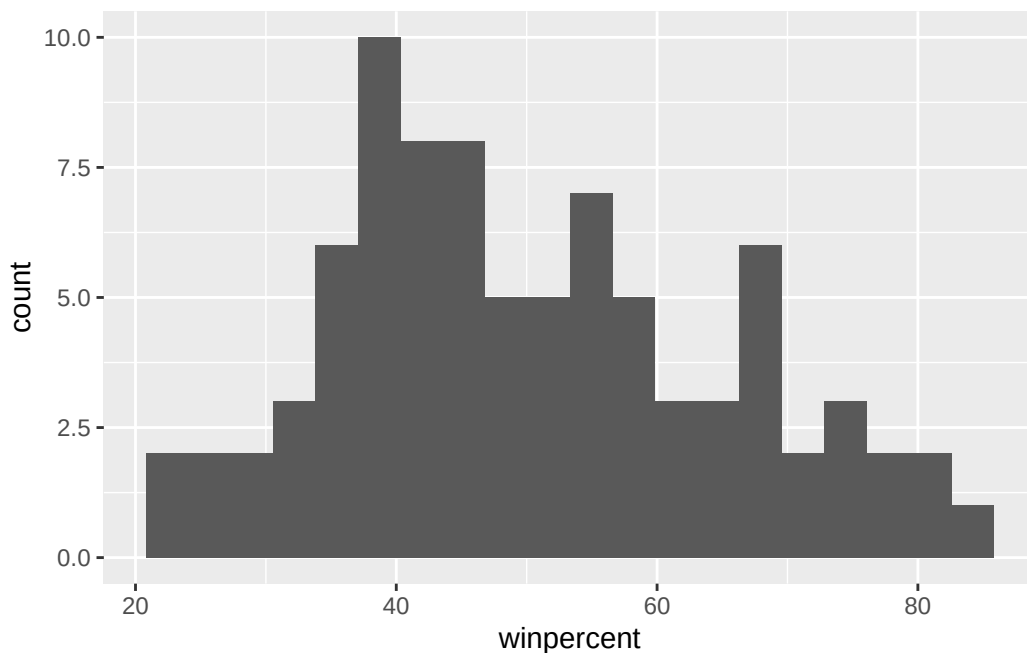
Q7. What do you think a zero and one represent for the candy\$chocolate column?

A zero would represent no chocolate, while a 1 would represent chocolate.

Q8. Plot a histogram of winpercent values

```
library(ggplot2)

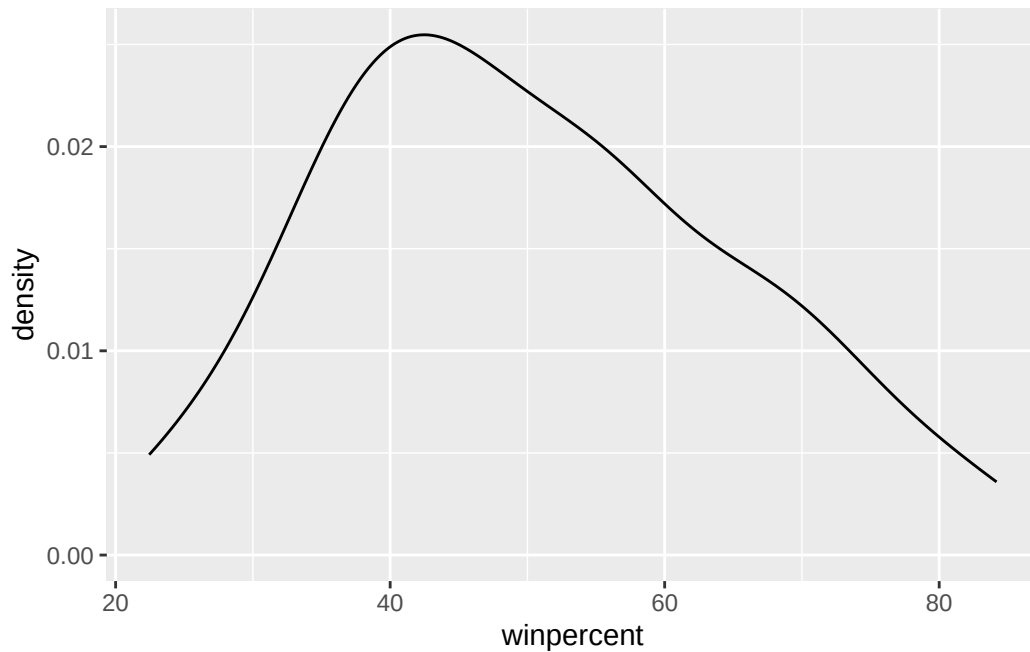
ggplot(candy)+
  aes(winpercent)+
  geom_histogram(bins = 20)
```



Q9. Is the distribution of winpercent values symmetrica?

The distribution is not symmetrical, it is skewed right.

```
ggplot(candy)+  
  aes(winpercent)+  
  geom_density()
```



Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

The center of the distribution is below 50 because the median is 47.83. This happens even if the mean is above the 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
# 1. Find all chocolate candy in the dataset
# 2. Find their winpercent values
# 3. Calculate the mean of these values

# 4-6. Do the same for fruity candy
# 7. Compare mean winpercents for chocolate vs. fruity

choc.inds <- candy$chocolate == 1
choc.win <- candy[choc.inds, ]$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```
frt.inds <- candy$fruity == 1
frt.win <- candy[frt.inds, ]$winpercent
mean(frt.win)
```

```
[1] 44.11974
```

Chocolate candy is on average ranked higher than fruity candy.

Q12. Is this difference statistically significant?

```
t.test(choc.win, frt.win)
```

Welch Two Sample t-test

```
data:  choc.win and frt.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

This means are statistically significant.

Q13. What are the five least liked candy types in this set?

```
candy |>
  arrange(winpercent)|>
  head(5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0

	crispedricewafer	hard	bar	pluribus	sugarpercent	pricepercent
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

```
candy |>
  arrange(-winpercent)|>
  head(5)
```

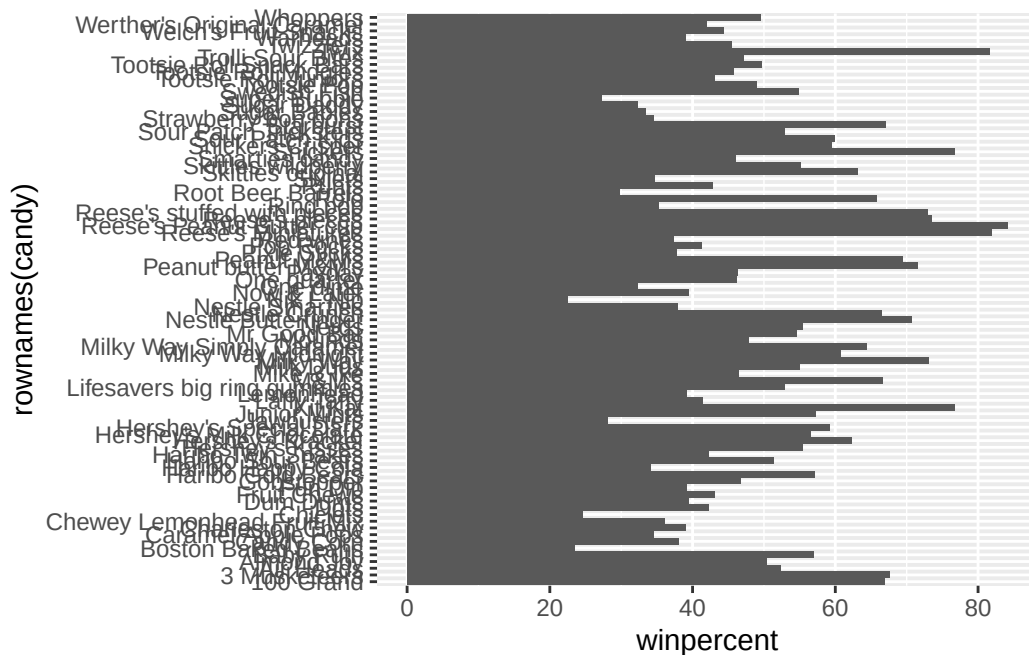
	chocolate	fruity	caramel	peanutyalmondy	nougat
Reese's Peanut Butter cup	1	0	0	1	0
Reese's Miniatures	1	0	0	1	0
Twix	1	0	1	0	0
Kit Kat	1	0	0	0	0
Snickers	1	0	1	1	1

	crispedricewafer	hard	bar	pluribus	sugarpercent
Reese's Peanut Butter cup	0	0	0	0	0.720
Reese's Miniatures	0	0	0	0	0.034
Twix	1	0	1	0	0.546
Kit Kat	1	0	1	0	0.313

Snickers	0	0	1	0	0.546
	pricepercent	winpercent			
Reese's Peanut Butter cup	0.651	84.18029			
Reese's Miniatures	0.279	81.86626			
Twix	0.906	81.64291			
Kit Kat	0.511	76.76860			
Snickers	0.651	76.67378			

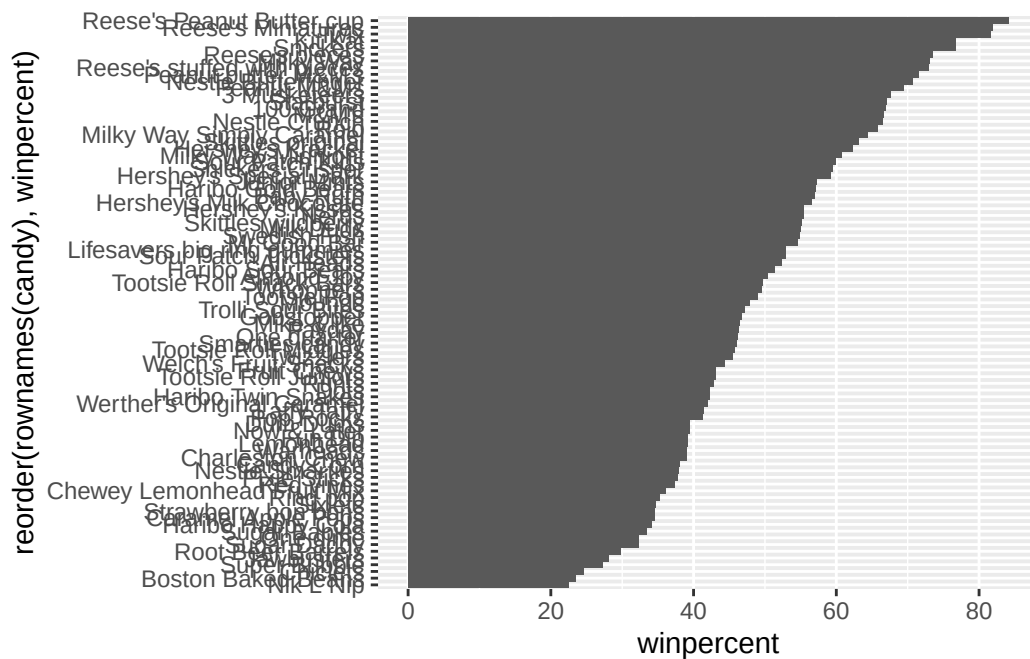
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```



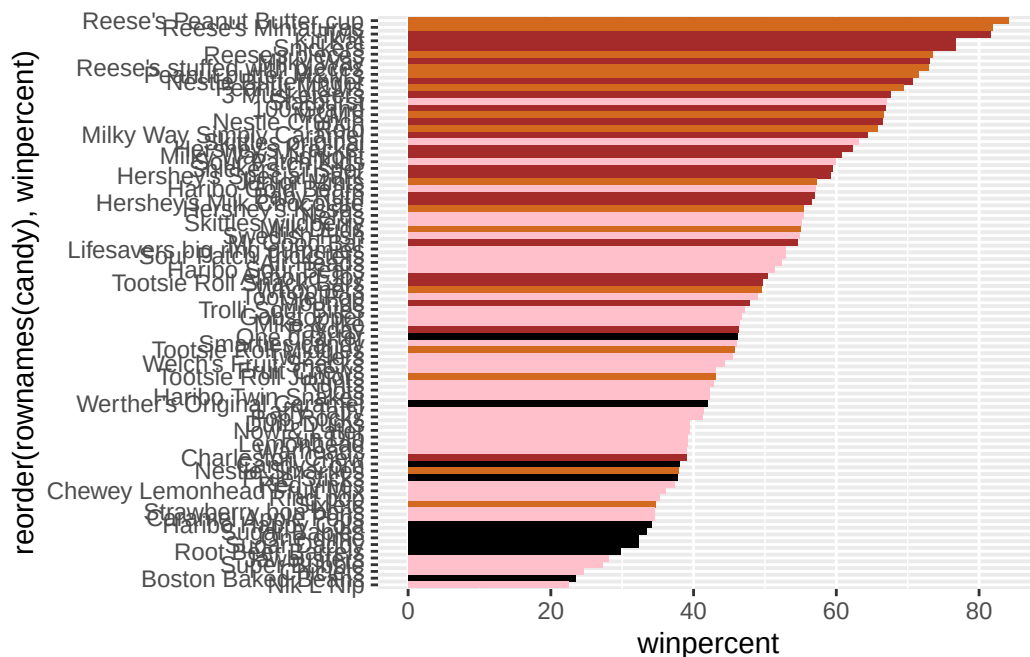
Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
ggplot(candy)+  
  aes(x=winpercent, y=reorder(rownames(candy), winpercent))+  
  geom_col()
```

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```



Q17. What is the worst ranked chocolate candy?

Sixlets are the worst ranked chocolate candy.

Q18. What is the best ranked fruity candy?

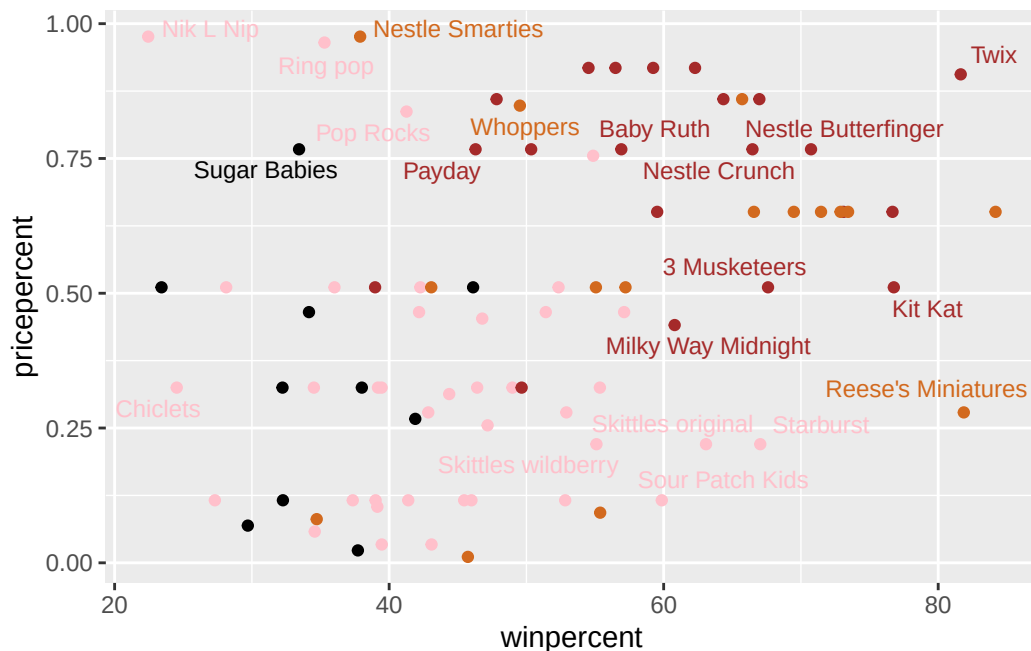
Starburst are the best ranked fruity candy.

Winpercent and Pricepercent

```
library(ggrepel)

# How about a plot of price vs win
ggplot(candy) +
  aes(winpercent, pricepercent, label= rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
candy |>
  mutate(bang_for_buck = winpercent / pricepercent) |>
  arrange(desc(bang_for_buck)) |>
  slice(1)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Tootsie Roll Midgies	1	0	0		0	0
	crisped	rice	wafer	hard bar	pluribus	sugarpercent
Tootsie Roll Midgies		0	0	0	1	0.174
	pricepercent	winpercent	bang_for_buck			
Tootsie Roll Midgies	0.011	45.73675	4157.886			

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
top5_expensive <- candy |>
  arrange(desc(pricepercent)) |>
  head(5)

top5_expensive
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Nestle Smarties	1	0	0	0	0
Ring pop	0	1	0	0	0
Hershey's Krackel	1	0	0	0	0
Hershey's Milk Chocolate	1	0	0	0	0

	crispedricewafer	hard	bar	pluribus	sugarpercent
Nik L Nip	0	0	0	1	0.197
Nestle Smarties	0	0	0	1	0.267
Ring pop	0	1	0	0	0.732
Hershey's Krackel	1	0	1	0	0.430
Hershey's Milk Chocolate	0	0	1	0	0.430

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

```
least_popular <- top5_expensive |>
  arrange(winpercent) |>
  slice(1)
```

```
least_popular
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer	hard
Nik L Nip	0	1	0	0	0	0	0

	bar	pluribus	sugarpercent	pricepercent	winpercent
Nik L Nip	0	1	0.197	0.976	22.44534

Exploring the correlation structure

```
candy <- candy[,-1]
cij <- cor(candy)
cij
```

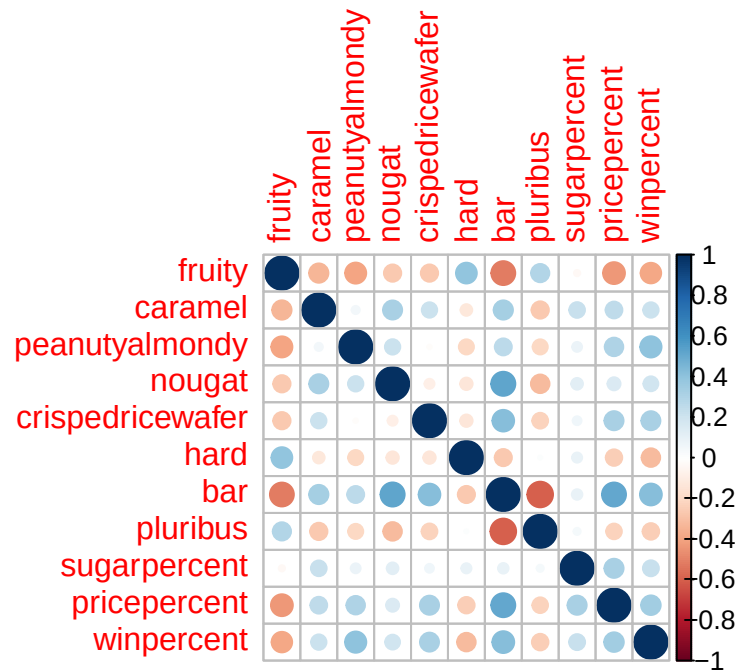
	fruity	caramel	peanutyalmondy	nougat
fruity	1.00000000	-0.33548538	-0.39928014	-0.26936712
caramel	-0.33548538	1.00000000	0.05935614	0.32849280

peanutyalmondy	-0.39928014	0.05935614	1.00000000	0.21311310
nougat	-0.26936712	0.32849280	0.21311310	1.00000000
crispedricewafer	-0.26936712	0.21311310	-0.01764631	-0.08974359
hard	0.39067750	-0.12235513	-0.20555661	-0.13867505
bar	-0.51506558	0.33396002	0.26041960	0.52297636
pluribus	0.29972522	-0.26958501	-0.20610932	-0.31033884
sugarpercent	-0.03439296	0.22193335	0.08788927	0.12308135
pricepercent	-0.43096853	0.25432709	0.30915323	0.15319643
winpercent	-0.38093814	0.21341630	0.40619220	0.19937530
	crispedricewafer	hard	bar	pluribus
fruity	-0.26936712	0.39067750	-0.51506558	0.29972522
caramel	0.21311310	-0.12235513	0.33396002	-0.26958501
peanutyalmondy	-0.01764631	-0.20555661	0.26041960	-0.20610932
nougat	-0.08974359	-0.13867505	0.52297636	-0.31033884
crispedricewafer	1.00000000	-0.13867505	0.42375093	-0.22469338
hard	-0.13867505	1.00000000	-0.26516504	0.01453172
bar	0.42375093	-0.26516504	1.00000000	-0.59340892
pluribus	-0.22469338	0.01453172	-0.59340892	1.00000000
sugarpercent	0.06994969	0.09180975	0.09998516	0.04552282
pricepercent	0.32826539	-0.24436534	0.51840654	-0.22079363
winpercent	0.32467965	-0.31038158	0.42992933	-0.24744787
	sugarpercent	pricepercent	winpercent	
fruity	-0.03439296	-0.4309685	-0.3809381	
caramel	0.22193335	0.2543271	0.2134163	
peanutyalmondy	0.08788927	0.3091532	0.4061922	
nougat	0.12308135	0.1531964	0.1993753	
crispedricewafer	0.06994969	0.3282654	0.3246797	
hard	0.09180975	-0.2443653	-0.3103816	
bar	0.09998516	0.5184065	0.4299293	
pluribus	0.04552282	-0.2207936	-0.2474479	
sugarpercent	1.00000000	0.3297064	0.2291507	
pricepercent	0.32970639	1.0000000	0.3453254	
winpercent	0.22915066	0.3453254	1.0000000	

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

```
which(cij == min(cij), arr.ind = TRUE)
```

```
      row col
pluribus  8  7
bar       7  8
```

The two most anti-correlated variables are pluribus and bar. This because they are represented in red and are the most negative values in the plot

Q23. Similarly, what two variables are most positively correlated?

```
cij_no_diag <- cij
diag(cij_no_diag) <- NA
which(cij_no_diag == max(cij_no_diag, na.rm = TRUE), arr.ind = TRUE)
```

```
      row col
bar     7  4
nougat  4  7
```

The most correlated variables are bar and nougat because they are represented in blue meaning that they are the most positive values.

Principal Component Analysis

The function to use is called `prcomp()` with an optional `scale=T/F` argument.

```
candy <- candy[,-1]
pca <- prcomp(candy, scale = T)
summary(pca)
```

Importance of components:

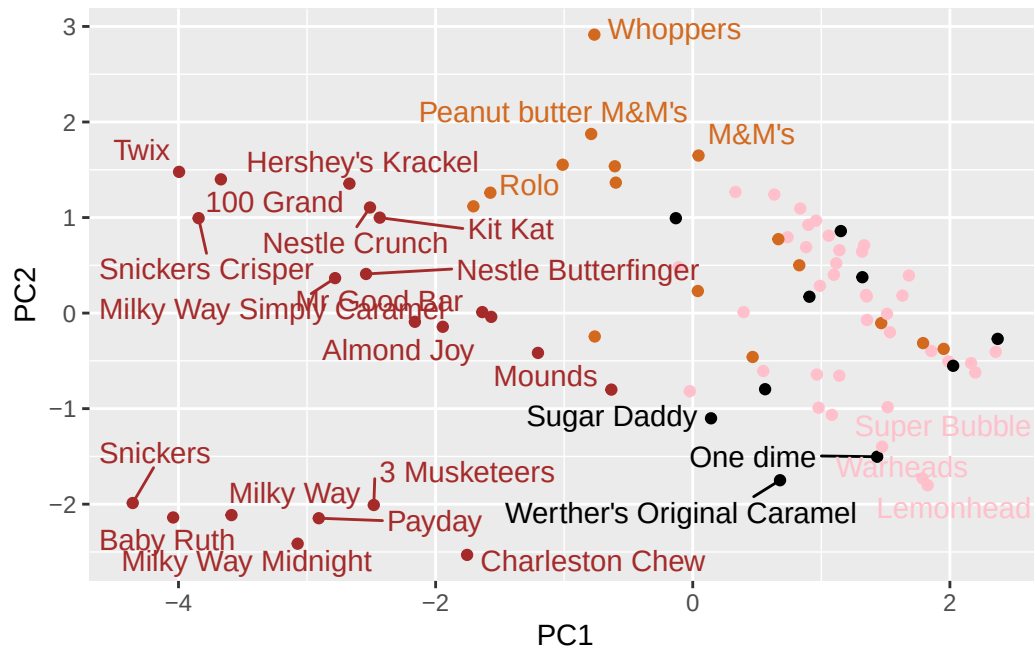
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	1.7995	1.1124	1.0862	1.0751	0.94440	0.81720	0.77601
Proportion of Variance	0.3238	0.1237	0.1180	0.1156	0.08919	0.06678	0.06022
Cumulative Proportion	0.3238	0.4476	0.5655	0.6811	0.77031	0.83709	0.89731

	PC8	PC9	PC10
Standard deviation	0.67812	0.60964	0.44206
Proportion of Variance	0.04599	0.03717	0.01954
Cumulative Proportion	0.94329	0.98046	1.00000

Our main PCA result figure

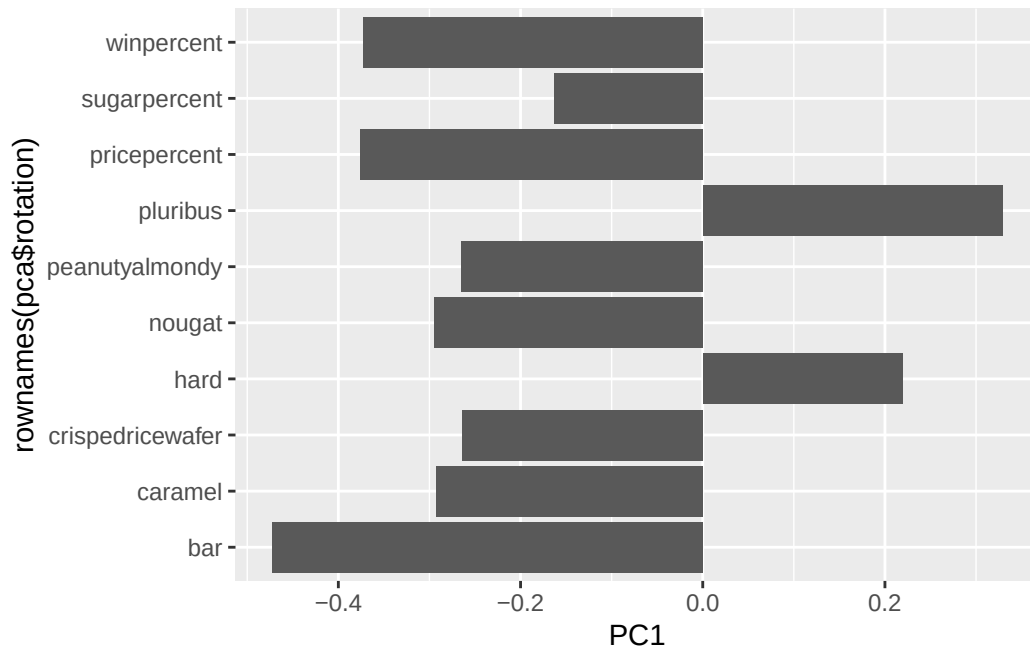
```
ggplot(pca$x) +
  aes(PC1, PC2) +
  geom_point(col = my_cols)+
  geom_text_repel(col = my_cols, label = rownames(candy))
```

Warning: ggrepel: 57 unlabeled data points (too many overlaps). Consider increasing `max.overlaps`



We should also examine the variable “loadings” or contributions of the original variables to the new PCs

```
ggplot(pca$rotation) +
  aes(PC1, rownames(pca$rotation)) +
  geom_col()
```

```
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

last_plot

The following objects are masked from 'package:flxtable':

highlight, style

The following object is masked from 'package:stats':

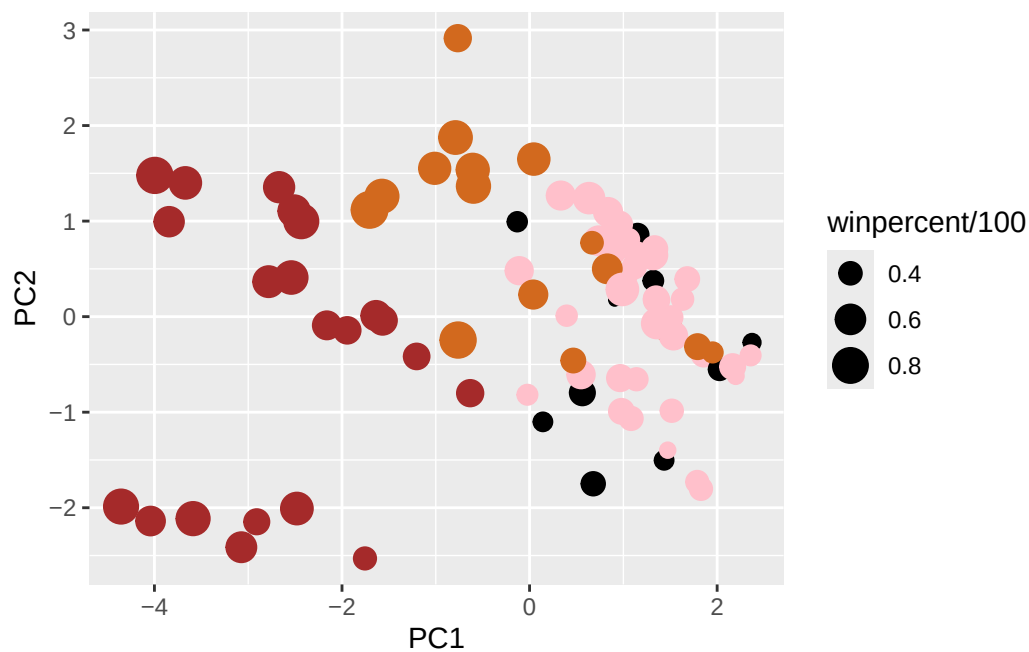
filter

The following object is masked from 'package:graphics':

layout

```
my_data <- cbind(candy, pca$x[,1:3])
p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=my_cols)
```

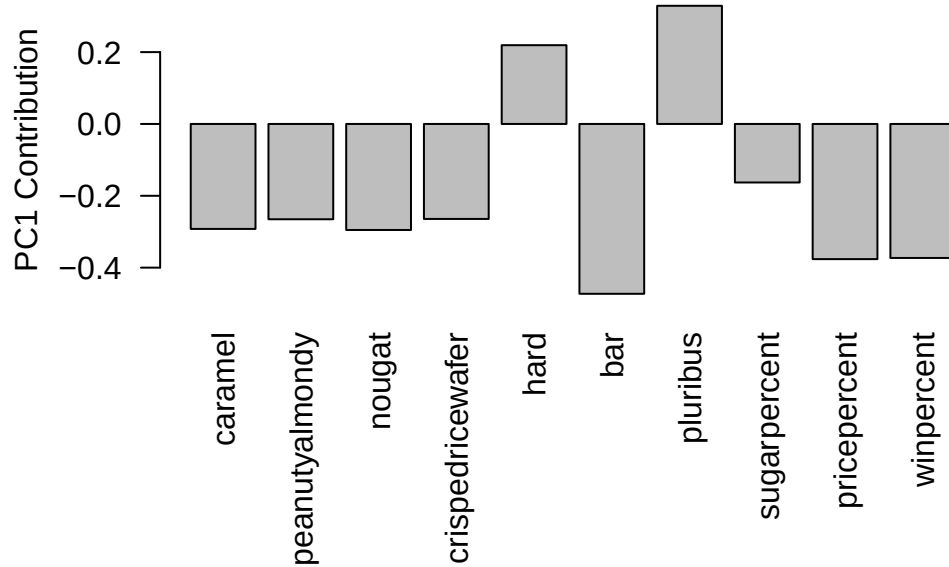
p



Interactive plots that can be zoomed on and “brushed” over can be made with the **plotly** package. It’s output is interactive and will not render to PDF :(.

```
#plotly(p)
```

```
par(mar=c(8,4,2,2))
barplot(pca$rotation[,1], las=2, ylab="PC1 Contribution")
```



Q24. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you?

The variables that are picked up strongly by PC1 in the positive direction are pluribus and hard. It kind of makes sense to me because it groups hard and pluribus candies together, and in a certain way when imagining a bag with a lot of candy I imagine mass production candies that are normally hard as well.